

“HINDHIVAC”
“OIL DIFFUSION PUMP”

OPERATION
AND
MAINTENANCE MANUAL



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The following warning symbols appear on the pump:



Warning – Risk of Electric Shock



Warning – Hot Surfaces



Warning – Personal Injury



PE Protective Earth

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1.0 INTRODUCTION

The Hindhivac Oil Diffusion Pumps are being manufactured in an extensive range to suit the varied needs of both industrial and laboratory users. These pumps are available with nominal bore sizes from a range of 1" to 36" diameter. While 1" pump gives 10 liters per second pumping speed the 36" pump gives 40 – 45000 lit/sec. with two to five stages (depending on size).

These water cooled Oil Diffusion Pumps are fabricated either from Mild Steel or Electroplated Aluminum or Stainless Steel. Two types of pumps are offered.

- a. Fractionating diffusion pumps and
- b. Non-fractionating pumps

The superior pumping speed of the HINDHIVAC Diffusion Pumps with other features make them an excellent choice of such high vacuum applications as Vacuum Furnaces, Space Research, Vacuum coating of lenses, thin film deposition and several other industrial applications.

The HINDHIVAC Diffusion Pumps produce much higher ultimate vacuum than any other type of vapour pumps like vapour ejector pump etc. Diffusion Pumps are employed in the pressure range 10^{-1} to 10^{-7} m.bar and even lower pressure using the correct technique.

2.0 GENERAL DESCRIPTION

2.1 MAIN BODY

The diffusion pump body is cylindrical in shape which has a housing for the jet assembly and the charging fluid. This is provided with a metallic flange at the top of which is either made of stainless steel or MS electro plated with suitable 'O' ring groove and securing bolt holes. This may be either fitted to the bottom flange or a suitable baffle valve or to the system to be evacuated.

Generally, all the pumps are fitted with a baffle valve or other type of baffles at the opening to avoid back streaming i.e., the migration of working fluid molecules into the system which ultimately lead to complete loss of the working fluid and contamination of the whole system.

Cooling water coil is soldered on the outer wall of the pump to cool the pump body.

2.2 JET

This consists of a set of metallic nozzles which are assembled around a central rod and housed in the main body of the pump. All the nozzles are oriented downwards at a certain angle towards the water cooled pump body and designed to give the best performance and maximum efficiency. The water circulated on the outer wall of the main body cools the vapour molecules which emerge from these nozzles striking the inner wall of the pump body.

Jet may be of a single or multi-stage depending on the size of the pump. In multi-stage the number of nozzles will be more than one. The jet accessories are made of stainless steel or electroplated aluminum. Diffusion pumps are manufactured in two types called the Standard Diffusion pump and Fractionating Diffusion pump. The main difference between these two is in the fabrication and arrangement of the nozzles. The working of these two explained in the section "Working Principles" separately.

2.3 BOILER

This is a part of the main body where the working fluid is heated and evaporated. The boiler is designed to hold a certain specified quantity of the recommended working fluid to give maximum efficiency. Always care should be taken to see that working fluid is not exposed to atmosphere when in hot condition. Also the level of the working fluid in the boiler housing should be often checked and kept constant. The boiler is heated electrically with a heating element housed outside the main body.

2.4 HEATERS

This is housed at the bottom of the main body of the pump. The heating element is properly secured to the main body by a suitable arrangement. The heating coil is well insulated from the main body and the heater cover. The terminals are provided on the other side of the heater cover. The terminals are provided on the outer side of the heater cover to connect to the main power supply. So always only the specified input has to be fed to the heating coils. Any variation in the heater input due to voltage variation will affect the performance of the pump.

2.5 CARTRIDGE HEATERS

In Modules with large diameters of diffusion pumps, cartridge heaters are used. In these pumps a burst free evaporation of the pump fluid is achieved by special heater construction and so, a pumping speed constant with time is obtained.

In this heating is achieved by an internal heater and consists of heater cartridges which are slid into tubes brazed conducting fins. The Stainless Steel tubes are welded horizontally in the pump body and located above the oil surface. The heat conducting fins which projects into the pump fluid is so chosen that an intensive but smooth evaporation of the pump fluid is obtained. The pump fluid receives additional energy from the parts of the conducting fins which stands above the surface. This special construction of the heater arrangement allows changing of the heater cartridges with pump still hot.

2.6 REPLACEMENT OF CARTRIDGE HEATERS

The cartridge heaters are made of either SS or brass sheathing which are ground on the outer surface to snugly fit in to the SS pipes welded to the pump body. Before inserting the cartridge heaters into the pipes clean inside surface of the pipes thoroughly and if necessary remove any burrs and dents with emery paper. Insert the heater completely into the pipe till it touches the covered end without any slackness. The electrical connections for the cartridge heaters are connected

to 3 phase supply with each phase connected to the three heaters separately and neutral is given to the other terminals of the three heaters together. Details of the connections are given in the circuit diagram enclosed with this working instruction.

2.7 FORM ARM

The fore-vacuum connection also known as backing connection diffusion pump which is situated at the side of the pump body, a little above the heater cover will have a flange or coupling. This is connected to some type of backing pump usually a rotary vacuum pump. It is normally directed upwards and is cooled by water to avoid pump fluid being discharged to the atmosphere by the backing pump. Baffles are provided inside the fore-arm to avoid further loss of fluid.

2.8 COLD CAP RINGS

Cold cap baffles are designed in such a way that they can be mounted inside the pump body directly above the top jet of the high vacuum stage. Cold top jet baffles are used to reduce the oil back streaming through the high thermal conductivity metallic spider legs. The cold cap has good thermal contact with the cooled wall of the pump which means it is practically at the cooling water temperature or the room temperature.

2.9 THERMOSTAT

This is a safety device which is fitted on the side of the main body above the cooling coils. This acts as a safety device to cut off the power supply of the boiler heater in case the cooling water supply fails or the temperature of the body and the fluid exceeds a safe limit. The working principle is explained in the section "Safety devices and Accessories".

3.0 WORKING PRINCIPLE

The working principle of both the Standard and the Fractionating type of Diffusion Pumps are almost the same. They differ only a little and the principle is as follows.

The oil in the boiler is heated by the heater and converted into vapour. This rises in the concentric columns and is limited by the jets due to the comparative high pressure existing above the boiler in the jet system. The vapour is forced through jet aperture where it is deflected downwards by the jet deflectors while the tubular side jet discharges vapour into the backing system. The molecules issuing from the jet engulf gas molecules, diffuse into the vapour streams not being able to diffuse back due to the downwards deflected vapours. The gas molecules are finally removed to the atmosphere by the backing pump. The oil vapour impinging on the water cooled pump wall condenses and drains to the boiler where it is re-evaporated.

In the conventional oil vapour pump the ultimate vacuum achieved is little less than the Fractionating type. This is due to the comparatively high volatility of the light fractions which arrive at the top jet and prevents achieving an ultimate vacuum lower than the vapour pressure of these fractions. Where as in the Fractionating type Diffusion Pump the condensed oil returning to the boiler for re-evaporation on its path to the centre of the boiler evaporates progressively. The light fractions or low molecular weight fractions of the high vapour pressure travel to the side jet as its temperature rises rapidly and evaporates before it reaches the centre of the jet. The medium fractions of little lower vapour pressure travels still further towards the centre but vapourises before they reach centre and feed the intermediate jets. Finally the oil composed mainly of the heaviest and the lowest vapour pressure fraction reaches the pump centre and vapourises, feeding the top jet. The characteristic which is termed "Fractionating" pumps thus allow only stable and heaviest fractions of the pump fluid to reach first stage which results in the high performance of the pump.

The pumps are available in single stage and multi-stages. The number of stages depends on the number of nozzle openings. HINDHIVAC pumps are available from two to five stages.

4.0 INSTALLATION

On receiving the equipment unpack it and remove all the packing materials. Remove the covers over the flanges and clean them carefully with acetone or some other suitable solvent without damaging the 'O' rings.

Pumps are dispatched with utmost care and if any damage is noticed in the jets or any others parts, they are to be immediately reported to us. All **HINDHIVAC** pumps are tested thoroughly for vacuum and high performance before it is dispatched to the customer.

The pump jets are fabricated out of SS or Aluminum. Precautions have to be taken while tightening the centre rod at the bottom with nut. Over tightening results in jet ends getting damaged which in turn will reduce the gap between the jets thus affecting the performance of the pump.

The **HINDHIVAC** pumps are supplied with fluid charge required for the diffusion pumps. If the pump is exposed to atmosphere before being installed or if a fluid other than the recommended oil is to be used the whole pump has to be cleaned with an organic solvent like acetone.

4.1 VACUUM TIGHTNESS

The pump should be tested for vacuum tightness before the fluid charge is placed into the boiler as the outgassing from the fluid may render the subsequent testing difficult. For most purposes it is sufficient to test the system with the rotary backing pump and assess the overall vacuum tightness by the closeness with which the vacuum attained at the inlet or at the apparatus.

4.2 PUMP FLUID

An ultimate vacuum in the range of 10^{-5} m.bar to 10^{-7} m.bar can be obtained with HINDHIVAC Diffusion pumps using Dow Corning 702, 704 and 705 silicon fluids. These are chemically inert and free from impurities. The 704 fluid is specially recommended for use along with liquid air traps to attain such low pressures as 10^{-7} m.bar

4.3 GENERAL PROPERTIES OF SILICON FLUIDS

Silicon fluids are not oxidized by air at operating temperatures. Their chemical resistance gives long life and eliminates frequent replacements, Silicon fluids being inert do not react with metal parts, Elastomer seals and gases such as Hydrogen and Carbon-Di-Oxide commonly found in high vacuum systems. Its radiation resistance is good because of high phenol contents. Silicon fluids will not break or decompose under operational conditions. In contrast all organic fluids undergo some cracking. Thermal stability fluids prevent decomposition.

There is no fouling of diffusion pump boiler surface and the jet with organic tar and hard carbon. Diffusion pumps using silicon fluids can operate against higher forepressures than those using most of other oils.

4.4 FLUID FILLING

It is recommended that only the oil charge specified by us are used to attain the maximum vacuum but other fluids of similar properties can be used. Whenever a different fluid is to be filled, the inner parts of the pump are to be cleaned thoroughly with acetone. On no account Mercury should be used.

The appropriate charge of oil specified in the instruction sheet for each pump is to be poured into the boiler. The charge may be poured down the inner wall of the pump body. Care must be taken that the fluid charge does not contact the upper stages of the jet assembly. The jet assembly is to be replaced properly after oil pumping as it was before taking enough care to see that the side opening of the bottom most jet should face the backing line operating provided in the main body. In case of Diffstak pump oil is filling through oil drain.

If the pump is fitted with an isolation baffle valve filling is down using a length of rubber or polythene tubing passed round the baffle plate when in the lifted position into the pump mouth. Insert tubing a little deep down into the body to avoid fluid coming into contact with top jets. The alternative method for charging the pump is by pouring oil into the backing opening after removing the vacuum union and its

flanges. Also it is possible to use the side drain lug-cum-oil level indicator by attaching a length of suitable rubber or diffusion pump thus sucking the fluid from the container into the boiler. This is possible in case of pump above 6" size. It is advised to check the oil level and the quality of fluid in the boiler periodically to obtain best performance. An easily accessible oil-cum-level checking device is fitted to all the pumps above 6" size.

4.5 VACUUM CONNECTION

The top flange of the pump may be secured to baffle valve (if used) or to any suitable mating flange of the system to be evacuated with a suitable 'O' ring groove. The backing connection provided on the side is connected to a suitable rotary pump with an ultimate vacuum better than the backing pressure required for the diffusion pump to operate. The coupling provided on the backing ejector pipe is coupled to a matching part of the coupling which is brazed to the pipe connecting the rotary pump. Care should be taken to clean the couplings with acetone and ensure that the 'O' ring is properly placed in the coupling. Pipe connecting the rotary pump to the backing connection should be of shortest possible length.

4.6 COOLING WATER AND CONNECTIONS

Connection should be made using flexible hoses to the cooling water pipe coupling provided so that the water flows downwards through the cooling coil. If a baffle valve is fitted the water should first flow through the baffle valve coil then through the vapour pump oil. The circulating water temperature should be kept as close as possible to 15°C. The recommended minimum water flow for various sizes of pumps and inlet temperatures are given in the chart.

The heater must not be switched on without the cooling water flowing, as the oil may be lost or decomposed and cooling coil solder may also melt.

For effective condensation of the oil vapours cold water at 15°C, cooled water should be circulated around the diffusion pump through the coils. It is essential

that the cooling water is dust free and tested for hardness to avoid formation of scales inside the cooling pipeline.

4.7 ELECTRICAL SUPPLY

The AC supply given to the pump heater should match the specified voltages for each pump mentioned in the chart. Connect the electric supply to pump heater via the two pin socket supplied along with the pump.

For OD-114 pump the power supply required for the heater will be of 1 phase 230V so that heater is connected to one phase as indicated in the circuit diagram enclosed. This will ensure proper voltage current supply to the pump heater for efficient heating. This can also be connected to 230V supply is given as indicated in the circuit diagram.

4.8 BACKING PUMP

A single stage or double stage rotary pump with a displacement not less than that is specified in the chart and capable of achieving a vacuum of 0.05m.bar or better is recommended and connected to the diffusion pump by means of shortest possible length of pipe.

5.0 OPERATING PRINCIPLE

(Assuming that it has to be operated in conjunction with a fully valved system).

STARTING

This applies when the pump and the apparatus are at the atmosphere pressure:-

- a. Close the high vacuum valve (baffle valve) air admittance valve of the pump as well as the chamber and all other openings connected to atmosphere.
- b. Start the rotary pump and then open the backing valve. After reaching 0.05m.bar or better pressure in Pirani gauge close the backing valve and open

the roughing valve connecting the chamber to the system. This ensures that both the backing and roughing lines are in vacuum.

- c. After ensuring that the backing pressure reaches 0.05m.bar, and cooling water supply is provided, switch on the diffusion pump.

Warning:

Diffusion pump should never be switched on without ensuring cooling water supply.

After a warm-up period of 20-30 minutes open the high vacuum valve (baffle valve) to connect the chamber to the diffusion pump.

Before opening the high vacuum valve, be sure that the chamber has minimum pressure 0.05 m.bar or better. As otherwise the pump will not be able to handle the volume efficiently.

6.0 CLOSING DOWN

Close the high vacuum or baffle valve and switch off the pump heater. Allow the pump to cool. Close the backing valve after sometime (roughly 15-20 minutes) and open the rotary pump air admittance valve and switch off the rotary pump. If magnetic isolation cum air admittance valve is installed at the inlet of the rotary pump the procedure of allowing air into the rotary pump is automatically down when the rotary vacuum pump is switched off. After the pump is completed cooled down, close the water supply.

NOTE:

This method ensure that the vapour pump is under vacuum thus preventing the pump fluid from absorbing air or decomposing due to cracking when air comes into contact with hot fluid.

NOTE:

Never allow air into the hot diffusion pump. If allowed the costly working fluid will be lost in decomposition.

7.0 SAFETY DEVICES OR ACCESSORIES

Thermostat or the thermal switch is an electrical device to protect the pump from damage due to failure in cooling water supply system. If the temperature of the pump rises too high, the heaters are automatically switched off. The thermostat is secured to a mounting plate which is attached to the pump cooling coil, and in the event of temperature exceeding the prescribed limit, power supply to the pump heater is cut-off the 5 amps and 230V AC rated switch should be connected in series with supply to the pump heater.

8.0 MAINTENANCE

FLUID CHANGE:

If the pump ceases to give satisfactory ultimate vacuum the fluid must be drained and inspected (See page fluid filling).

NOTE:

Remove the jet assembly by unscrewing the top jet or nut over it and remove the complete jet assembly, and slowly pour out the fluid into a container. If it is badly charred or discoloured the pump must be thoroughly cleaned and re-charged with new fluid.

IMPORTANT

1. Choose an installation area with suitable power outlet and ample water supply and drain. The area should be free from strong drafts and dust.
2. Check the pump to be sure that the jet assembly is aligned properly.

3. Fill the pump with the correct measure of recommended oil after thoroughly cleaning with acetone if exposed to dust and atmosphere for a long time.
4. Always mount the pump in a vertical position so as to have reasonably equal level of oil in the boiler.
5. Make tight vacuum connections on the high vacuum and fore vacuum sides. These will be generally flanges or couplings connection with suitable 'O' rings or gaskets. Connection to the backing pump should be preferably flexible to avoid detrimental effects due to vibration.
6. Check and match the water flow rate with the specified ratings, without any obstruction.
7. Verify the readiness of the safety devices like thermostat and water flow switch.
8. Switch on the heater only after the cooling water circulation is on and the required fore vacuum is reached.
9. Check whether the heater is working properly. This is usually done by feeling the boiler or heater cover.
10. The pressure should drop rapidly when the pump has reached its operating temperature. If not check the system for any probable leakage.
11. While shutting down the diffusion pump, the important precaution to be taken are to avoid letting air into the diffusion pump while the fluid is still hot so as to prevent the damage to the pump oil due to cracking. Even silicon oils is decomposed to some extent by exposure to air when hot.
12. Check the heater input supply for proper voltage and wattage as specified.

13. Change oil depending on the nature of vacuum process to which the pump is to be used.
14. Check the condition of the oil and the oil level occasionally.
15. Set up a cleaning programme depending on the nature of usage of pump.
16. Whenever the performance of the pump comes down, clean the pump thoroughly and refill with new oil.
17. Do not use cheap substitute of fluid charge. Use only the recommended oil or its equivalent.

ACCESSORIES TO DIFFUSION PUMPS

Selection of appropriate accessories should be based on the kind of application the system is intended to be deployed, to achieve efficient utilization of the system.

A simple pumping system consisting of a diffusion pump backed by a rotary pump attached directly to the vessel to be evacuated enables the most efficient use of the Available pumping speed. However the use of vapour diffusion pump of any type usually results in the presence to a considerable degree of the pump fluid into the system. This results in the loss of costly pumping fluids as well as contaminating the vacuum system and thus affecting the process involved.

The phenomena of the flight of the vapour molecules into the system from the pump inlet is called back streaming. The major source of back streaming is the top of the first jet.

Depending on the process to which vacuum is employed where high degree of cleanliness vapours is required a large variety of vacuum accessories for these

diffusion pumps are designed and manufactured by **HINDHIVAC PVT LTD** to meet the user's requirements.

In order to eliminate this back streaming to a large extent it is advisable to use a cooled optically dense baffle or a trap which acts as a barrier to the migrating fluid. **HINDHIVAC** pumps are designed to give the overall performance with minimum back streaming but where complete back streaming is to be avoided and oil contamination is to be reduced to the minimum, the customer should choose some of the following accessories which can be directly fitted to the diffusion pump flange.

BAFFLE VALVE

HINDHIVAC water cooled baffle valve enables a working vapour pump to be isolated while the pumping system is at atmospheric pressure. The valve plate just on top of the diffusion pump acts as an effective baffle in preventing the migration of oil vapours.

These baffles are made in two basic versions:

1. In-Line type
2. Right Angle type

For further details on this please refer to the brochure on "Baffle Valves".

CHEVRON BAFFLES

These are optically dense baffles which are compact with maximum conductance and with effective trapping of back streaming vapours.

We offer two types, the first being "Water Cooled Chevron Baffle" and the second is "Refrigerated Chevron Baffles".

These can be easily installed on top flange of the diffusion pump and is fitted with a butterfly valve at the top or a gate valve for isolating the working pump when the top vessel is to be opened to atmosphere.

When a better oil free chamber is required the introduction of a refrigerated chevron baffle is recommended for on top of the vapour pump.

BUTTERFLY VALVES

HINDHIVAC quarter swing type butterfly valves are compact and high conductance valves. Its compact dimensions help to reduce the height of the system. These are recommended for use as baffles in combination with a chevron baffle or with a liquid air trap.

LIQUID AIR/NITROGEN TRAPS

HINDHIVAC liquid air traps are optically dense and constructed out of 304 non-magnetic stainless steel. These are used to trap pump fluid vapours migrating to the system. They are designed for liquid gas coolants such as liquid nitrogen, Liquid air or solid CO₂ with alcohol mixture. Highly polished interior surface reduces heat loss as by radiation. The above stated are some of the essential accessories for obtaining an ultimate and contamination free vacuum in shortest possible time.

CRYO-COOLED LIQUID NITROGEN TRAP

A cryo-cooled liquid nitrogen trap is fixed at the side of the diffusion pump to avoid the back streaming and to achieve the clean ultimate vacuum.

This trap is giving good through put for vacuum pumping when compared with standard LN₂ trap. It occupies less height for better operational height of the unit. Without reworking the pipeline setup, the trap either can be removed or fixed. The trap has an OFHC Cu baffle which is very effective and avoids back streaming of oil.

13.0 TECHNICAL SPECIFICATIONS

PARAMETERS	OD-65	OD-80	OD-114D	OD-150D	OD-250	OD-250D	OD-350	OD-500
Nominal dia (mm)	65	80	114	150	250	250	350	500
Pumping Speed (lit/sec)	120	240	280	750	3000	1700	6000	12000
Backing Pump	ED-6	ED-6	ED-15	ED-21	CD-90	CD-45	CD-120	CP-250 + RA-375
Warm up time in Minutes	15	15	15	15	15	15	15	15
Ultimate Vacuum m.Bar (DC704)	5×10^{-7}	5×10^{-7}	5×10^{-7}	5×10^{-7}	5×10^{-7}	5×10^{-7}	5×10^{-7}	5×10^{-7}
Fluid Quantity (CC)	80	80	100	250	600-800	500	1200- 1800	3000-3500
Heater Rating (Watts)	350	350	500	1250	2250	2250	3750	7500
Material Body	SS	SS	SS	SS	SS	SS	SS	SS
Material Jet Assembly	SS	AL	AL	SS & AL	AL	SS & AL	AL	AL
Cooling Water required in Lits/Mins at 20-25 °C	2	AIR COOLED	2	4	5-6	4	6-7	6-8